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Together above two equations are called secular equations

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This could now be applied to a compound like ethene. We can divide each element by β and obtain

$$\begin{vmatrix} x & 1 \\ 1 & x \end{vmatrix} = 0; \text{ where } x = \frac{\alpha - E}{\beta} \implies E = \alpha - x\beta$$

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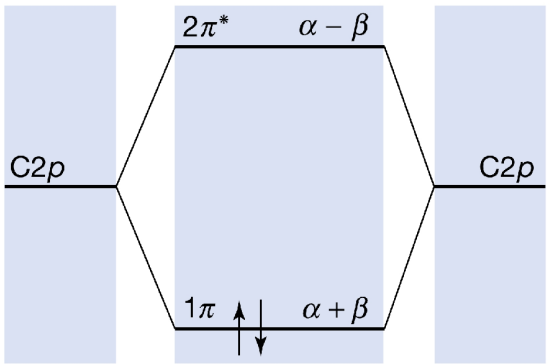
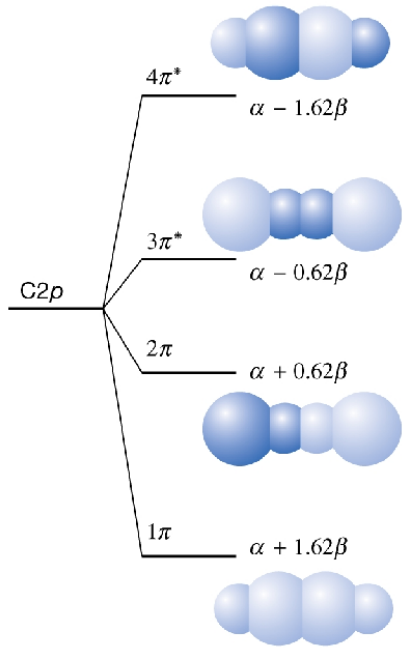
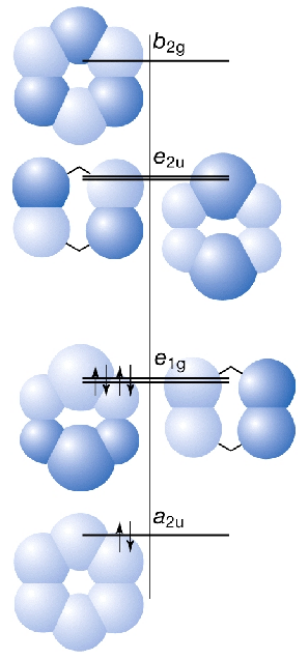
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3. For more then two atoms, we also ignore the β beyond the next neighbour.
For butadiene,

$$\begin{vmatrix} x & 1 & 0 & 0 \\ 1 & x & 1 & 0 \\ 0 & 1 & x & 1 \\ 0 & 0 & 1 & x \end{vmatrix} = 0$$

Ethylene	Butadiene	Benzene
		

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$$\Rightarrow x \begin{vmatrix} x & 1 & 0 & 0 & 0 \\ 1 & x & 1 & 0 & 0 \\ 0 & 1 & x & 1 & 0 \\ 0 & 0 & 1 & x & 1 \\ 0 & 0 & 0 & 1 & x \end{vmatrix} - \begin{vmatrix} 1 & 1 & 0 & 0 & 0 \\ 0 & x & 1 & 0 & 0 \\ 0 & 1 & x & 1 & 0 \\ 0 & 0 & 1 & x & 1 \\ 1 & 0 & 0 & 1 & x \end{vmatrix} + \begin{vmatrix} 1 & x & 1 & 0 & 0 \\ 0 & 1 & x & 1 & 0 \\ 0 & 0 & 1 & x & 1 \\ 0 & 0 & 0 & 1 & x \\ 1 & 0 & 0 & 0 & 1 \end{vmatrix} = 0$$

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we find an additional stability of 2β for benzene. This is the stability due to delocalization of the electron density over the skeleton.

Semiconductors:
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$$\begin{vmatrix} x & 1 & 0 & . & . & . & . & . & 0 & 0 & 0 \\ 1 & x & 1 & 0 & 0 & . & . & . & 0 & 0 & 0 \\ 0 & 1 & x & 1 & 0 & . & . & . & 0 & 0 & 0 \\ 0 & 0 & 1 & x & 1 & . & . & . & 0 & 0 & 0 \\ . & . & . & . & . & . & . & . & . & . & . \\ . & . & . & . & . & . & . & . & . & . & . \\ . & . & . & . & . & . & . & . & . & . & . \\ 0 & 0 & 0 & 0 & 0 & . & . & . & 1 & x & 1 \\ 0 & 0 & 0 & 0 & 0 & . & . & . & 0 & 1 & x \end{vmatrix} = 0$$

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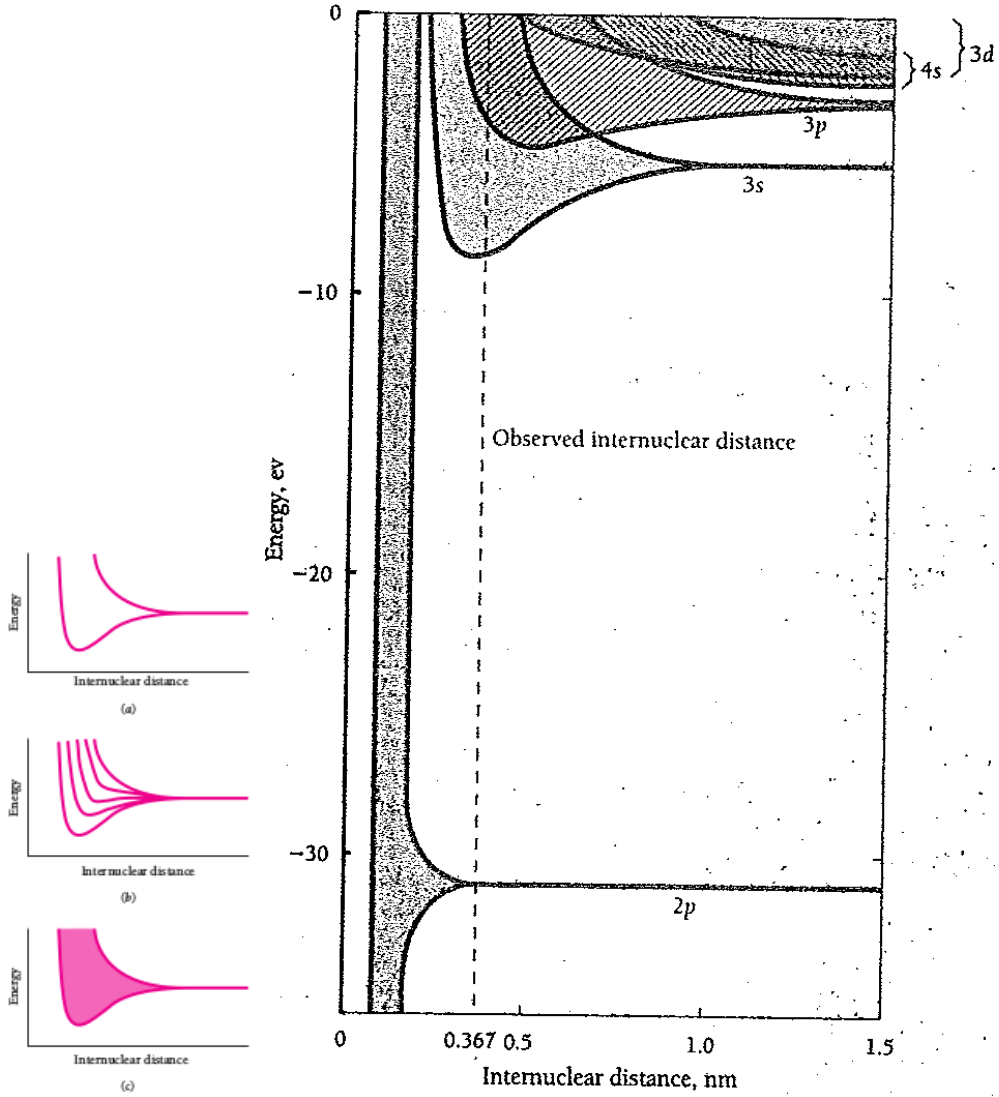
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all N MOs formed are stacked between energy interval of 4β. For p-orbitals of carbon atoms, $\beta = 75$ kJ/mol. Thus, we have $N = O(10^{24})$ MOs stacked in an energy interval of 300 kJ/mol. This is how we have band formation. Different bands formed from he atomic orbitals may or may not overlap at given internuclear distances.

following two pictures from Beiser's text book show how band formation takes place in sodium.



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figure below shows how carbon is an insulator, because at its equilibrium internuclear distance of ~ 1.5Å, the gap between CB and VB is 6eV and silicon is an insulator, because at its equilibrium internuclear distance, the gap between CB and VB is 1eV

